Report

GRAVITATIONAL WAVES

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INTRODUCTION

In 2015 gravitational waves were detected for the first time by the LIGO-Virgo collaboration. This triumph happened 100 years after Albert Einstein's formulation of the Theory of General Relativity and 99 years after his prediction of gravitational waves.

In General Relativity, the universe is described as a spacetime manifold with a curved metric whose curvature encodes the properties of the gravitational field. While sometimes one wants to use General Relativity to describe the whole universe, often we just want to know how a single object or small collection of objects behaves. To address that kind of problem, we use the idealization of the isolated system: a spacetime consisting of just the objects we want to study and nothing else. We might consider the solar system as an isolated object, or a pair of black holes spiraling into one another until they collide. We ask how those objects look to a distant, far away observer in a region where presumably the curvature of spacetime is very small. Gravitational waves are vibrations in spacetime that propagate at the speed of light away from their source. They may be produced, for example, when black holes merge. This is what was first detected by Advanced LIGO (aLIGO) and this is the focus of this article.

We conclude with the experimental detection of gravitational waves and the astrophysical implications of this detection. This detection is not only a spectacular confirmation of Einstein's theory, but also the beginning of the era of gravitational wave astronomy, the use of gravitational waves to investigate aspects of our universe that have been inaccessible to telescopes.

WHAT ARE GRAVITATIONAL WAVES

Gravitational waves are ripples in the fabric of spacetime caused by the acceleration of massive objects, such as merging black holes or neutron stars. These waves propagate outward from their source at the speed of light and carry energy across the universe. They were first predicted by Albert

Einstein in 1915 as part of his General Theory of Relativity, which describes how mass and energy warp spacetime.

FORMATION

When massive objects move or collide, they create disturbances in spacetime, much like how a stone creates ripples on the surface of water. These waves stretch and compress space as they pass through it, affecting the distance between objects.

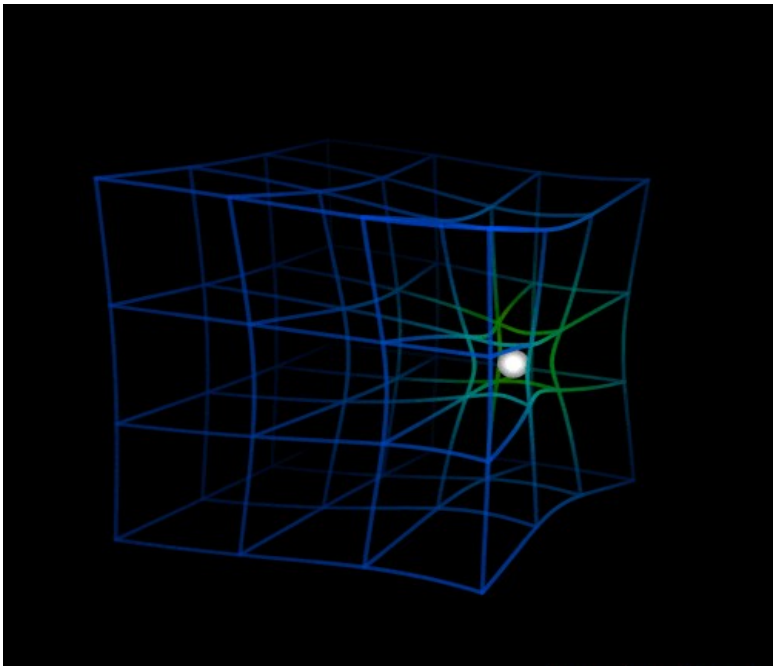

Fig.:-Representation of a stellar body warping the space-time fabric. https://preview.redd.it/ntiyg1fkvth41.gif?format=png8&s=c96b88b7a8cb877c5f22d1555911ec2cab4cd2ee

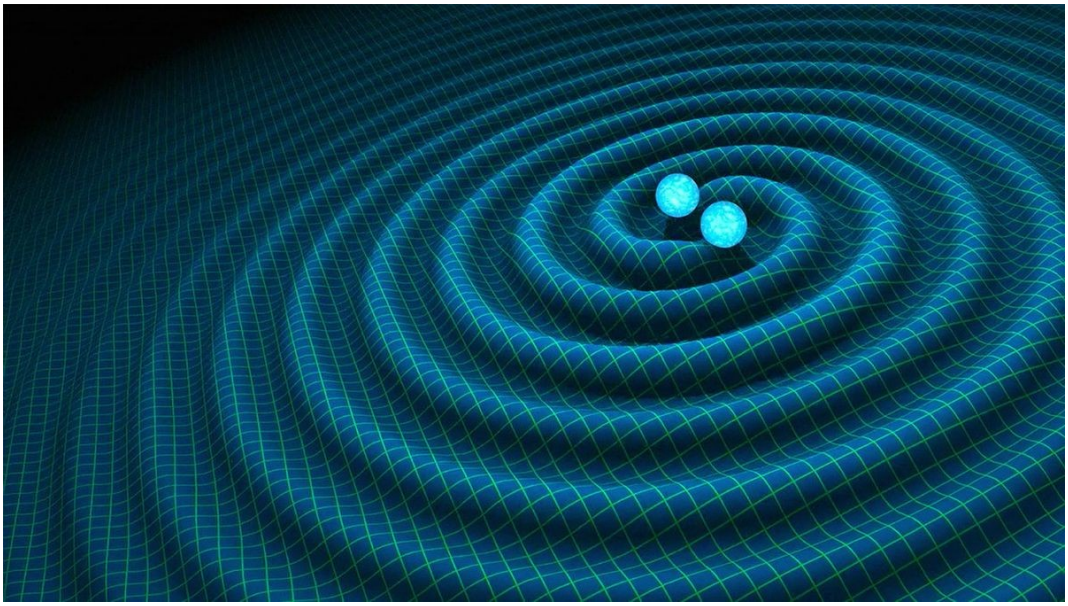

Fig.:- A system of binaries producing gravitational waves by creating ripples in space-time fabric. https://cdn.mos.cms.futurecdn.net/ydwsrwtqwfCre2v6KPpJcn-1200-80.jpg

Gravitational waves are extremely faint by the time they reach Earth, making them very difficult to detect. The first direct detection of gravitational waves was made in 2015 by the Laser Interferometer Gravitational-Wave Observatory (LIGO), confirming Einstein's predictions.

FIRST DETECTION OF GRAVITATIONAL WAVES

The first-ever detection of gravitational waves occurred on September 14, 2015, by the Laser Interferometer Gravitational-Wave Observatory (LIGO). This groundbreaking event confirmed a major prediction made by Albert Einstein's General Theory of Relativity a century earlier. The detection marked the birth of a new field in astronomy—gravitational wave astronomy—and was one of the most significant scientific discoveries of the 21st century.

What Happened During the First Detection?

The gravitational waves detected on September 14, 2015, were produced by the merger of two black holes. These waves came from the collision and merger of two black holes, each about 30 times the mass of the Sun. These black holes were located about 1.3 billion light-years away from Earth, in a distant part of the universe. As the two black holes spiraled toward each other, they emitted intense gravitational waves, which propagated outward at the speed of light.

As the black holes orbited each other, their immense masses created disturbances in the fabric of spacetime. These ripples stretched and compressed space itself, causing tiny shifts in distance between objects on Earth, detectable by extremely sensitive instruments.

LIGO, an observatory designed to detect these tiny ripples in spacetime, consists of two large detectors—one in Hanford, Washington, and the other in Livingston, Louisiana. The detectors use laser interferometry, where a laser beam is split into two arms that travel at right angles to each other. When a gravitational wave passes through, it slightly changes the length of the arms, which can be measured with incredible precision.

In the case of the September 14 detection (known as GW150914), the signal was so strong that it was detected simultaneously at both LIGO sites, confirming the wave's authenticity.

The gravitational wave signal was recorded for about 0.2 seconds, during which time the black holes merged and released an enormous burst of energy. The signal began with a "chirp" (a sharp increase in frequency and amplitude) as the black holes accelerated towards each other. This "chirp" is a hallmark of a binary black hole merger. The frequency of the gravitational waves increased as the black holes spiraled inwards, and then sharply peaked when they merged. The wave then tapered off after the merger. The event was dubbed **GW150914**, after the date of the detection.

The gravitational waves caused a change in the distance between the arms of the detectors by less than one-thousandth of the diameter of a proton. Despite this minuscule change, LIGO’s instruments were able to detect it with an unprecedented level of sensitivity. These waves are difficult to observe as unlike electromagnetic waves which propagates in the framework of space-time, gravitational waves are waves of the spacetime fabric itself. Also, the magnitude of gravitational waves is inversely proportional to the separation of the source and observing point, hence, it gets fainted reaching earth.

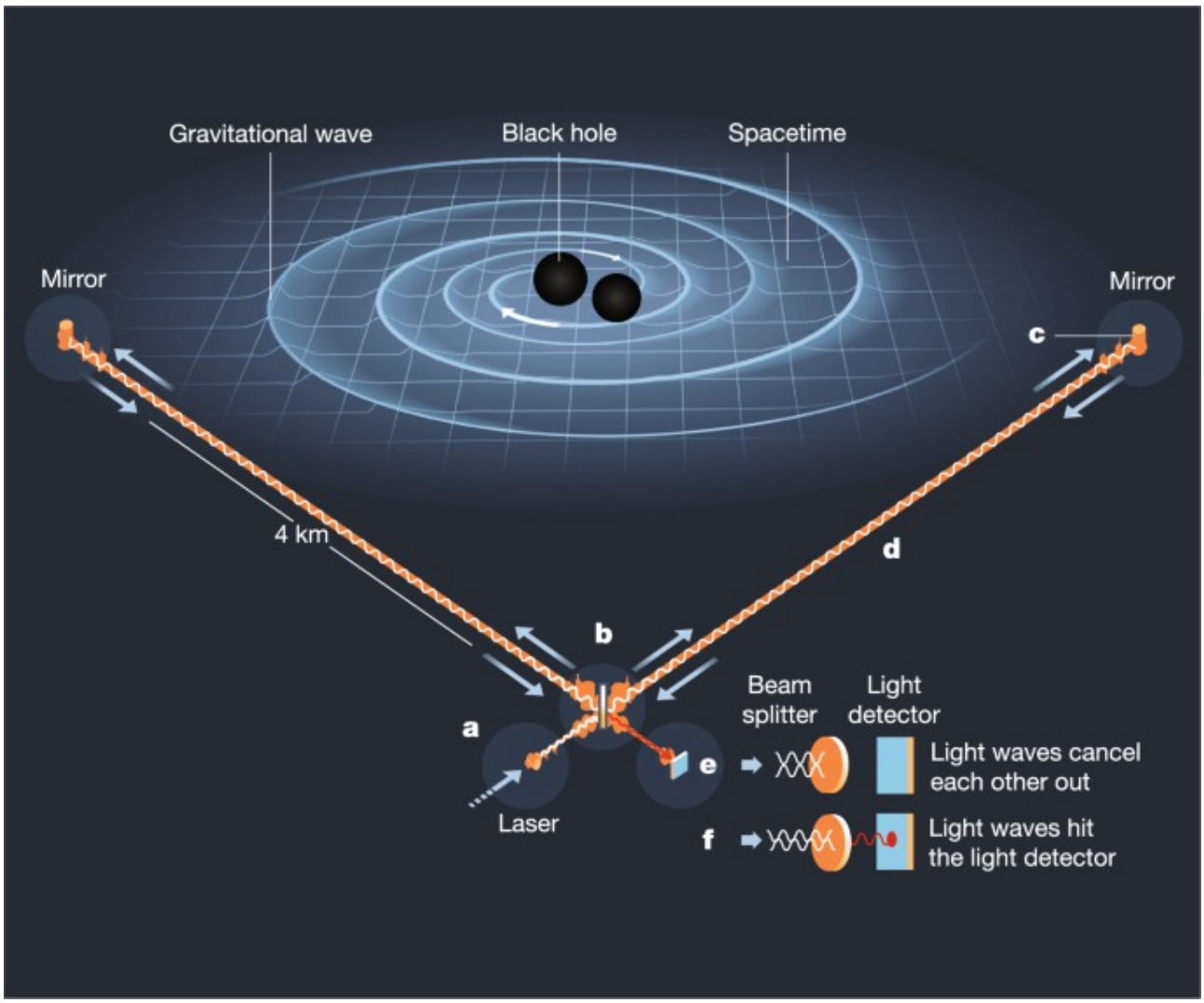

Fig.:- Representation of first ever detection of Gravitational waves.
https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2Fs41586-019-1129-z/MediaObjects/41586_2019_1129_Fig1_HTML.png

SIGNIFICANCE OF THE DISCOVERY

1. Confirmation of Einstein’s Theory:

   The detection of gravitational waves provided direct confirmation of Einstein's General Theory of Relativity. Einstein had predicted the existence of gravitational waves in 1915, but their direct detection was a major scientific challenge due to their extremely faint nature. The detection of GW150914 was considered a spectacular validation of his century-old predictions about the nature of gravity.

2. New Window into the Universe:

   Gravitational waves opened a new way to observe the universe. Unlike electromagnetic waves (light, radio waves, etc.), gravitational waves can pass through matter without being absorbed or scattered. This means they provide a direct view of objects and events that are otherwise invisible to traditional telescopes, such as black hole mergers and neutron star collisions.

   The ability to "hear" the universe in this new way enables scientists to study extreme environments like the interiors of black holes, where no other observations can be made.

3. A New Era in Astronomy:

   This discovery heralded the beginning of gravitational wave astronomy, an entirely new field of research. Following this detection, more gravitational wave events were observed, allowing scientists to learn about exotic astrophysical phenomena in unprecedented detail.

   It also set the stage for future space-based detectors, such as the Laser Interferometer Space Antenna (LISA), which is planned to detect lower-frequency gravitational waves from massive objects like supermassive black hole mergers.

4. Nobel Prize in Physics:

   In 2017, the Nobel Prize in Physics was awarded to Rainer Weiss, Barry C. Barish, and Kip S. Thorne, the key scientists behind the LIGO project, for their decisive contributions to the detection of gravitational waves.

CONCLUSION

The first detection of gravitational waves (GW150914) was a monumental achievement in physics and astronomy. It confirmed the existence of gravitational waves, validated Einstein's theory of general relativity, and opened a new observational window to the universe. By detecting the gravitational waves from merging black holes, scientists were able to directly observe some of the most extreme events in the cosmos, leading to a revolution in our understanding of the universe.

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